Wenyue Hua

Senior Researcher & Technical Lead, Microsoft Research, AI Frontiers

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Google Scholar • GitHub • OpenReview

Research Areas

- **Human-centered AI**: computational social science, social simulation, and AI applications in humanities and finance.
- **Trustworthy agentic AI**: agent safety, privacy-preserving alignment, insurance and risk modeling for agents
- **Efficient agentic systems**: agentic request scheduling, speculative planning, model routing and selection
- **Multi-agent systems**: multi-agent negotiation, persuasion, adversarial attack for multi-agent systems, large-scale agentic simulation

Selected Leadership and Impact

- **Frontier on-device post-training**: technical lead for a 4B on-device model training spanning six persuasion and negotiation domains. Built an environment- and harness-agnostic proxy-server training platform and the SFT/PPO pipeline; cascade RL **outperformed GPT-5.2** on the consolidated six-domain evaluation, while multi-teacher on-policy distillation unified six specialist teachers into one deployable model retaining **94% of teacher accuracy**.
- **Production-scale multi-agent systems**: Integrate the research of **multi-user, multi-agent Calendar Scheduling** initiative of the above 4B-model post-training to Calendar product adopted both by company internal usage as well as external product with massive cost reduction.
- **Research that became product**: built and led the open-source **AgentOpt** system from research prototype to adoption in **Microsoft Foundry**, establishing a framework-agnostic model-selection and routing layer for agentic workflows. Across AgentOpt and two speculative planning systems, delivered **40% lower end-to-end planning latency**, **30% lower total cost**, and **62-76% lower model-selection evaluation budget**.
- **Field-building and public influence**: research featured in *Fortune*, with the author quoted on systemic financial risk from AI agents; co-organizer of the ICLR 2026 MemAgents and COLM 2026 Lifelong Agents workshops; invited seminars at Columbia, Carnegie Mellon, UC Santa Barbara, MBZUAI, and Fermilab.

Full-Time Appointments

Microsoft Research, AI Frontiers — Jun. 2025 – Present
Senior Researcher & Technical Lead, Agent Post-Training for Multi-agent Negotiation — *New York, NY*

- Lead the research program for **multi-user, multi-agent Calendar Scheduling** within a product team, defining model-training, multi-agent RL, privacy-alignment, and evaluation strategies for reliable coordination among users and agents.
- Serve as **technical lead for post-training 4B on-device persuasion and negotiation models**; built an environment- and harness-agnostic proxy-server training infrastructure that decouples training from inference.
- Trained models with SFT and PPO across six domains, each reaching frontier-model-level capability; developed transfer-aware consolidation recipes through cascade RL and multi-teacher on-policy distillation, surpassing GPT-5.2 and retaining 94% of teacher accuracy, respectively.
- Lead the open-source **AgentOpt** repository; following strong research results, Microsoft adopted and integrated AgentOpt into Foundry for cost-efficient model selection in agentic workflows. UCB-E search recovers near-optimal assignments with 62–76% less evaluation than brute force.

University of California, Santa Barbara — Oct. 2024 – Jun. 2025
Postdoctoral Researcher, Department of Computer Science — *Santa Barbara, CA*

- Co-hosted by Carnegie Mellon University, Computer Science department, with Prof. Fei Fang and Prof. Hong Shen.
- Conducted research on efficient and trustworthy AI agents and LLM reasoning with Prof. William Yang Wang.
- Led or co-led work on inductive reasoning evaluation, semantic scheduling, dynamic speculative planning; Dynamic Speculative Planning reduced total cost by 30% while preserving lossless acceleration.

Education

Rutgers University-New Brunswick, Ph.D. in Computer Science — 2024
Dissertation: Trustworthy Large Language Models; advisor: Prof. Yongfeng Zhang

Rutgers University-New Brunswick, M.A. in Linguistics — 2020
Thesis: Learning Underlying Representations and Input-Strictly-Local Functions; advisor: Prof. Adam Jardine

University of California, Los Angeles, B.S. in Mathematics; B.A. in Linguistics and Philosophy, Specialization in Computing — 2018
GPA: 3.8/4.0; advisor: Prof. Edward Keenan

Student Mentorship and Teaching

- **Graduate research mentorship**: directs graduate research across three institutions, providing end-to-end guidance from problem formulation and experimental design through publication. Resulting peer-reviewed work includes:
  - **Yilin Guan** (M.S., Johns Hopkins University): *Dynamic Speculative Agent Planning* — **ICLR 2026**.
  - **Ruiwen Zhou** (Ph.D., National University of Singapore): *RuleArena: A Benchmark for Rule-Guided Reasoning with LLMs in Real-World Scenarios* — **ACL 2025**.
  - **Kevin (Kangqi) Ni** (Ph.D., University of North Carolina, Chapel Hill): *AdaSpec: Adaptive Routing by Ranking for Speculative Agents* — **EMNLP 2026**.
- **Teaching**: teaching assistant for *Natural Language Processing* (graduate), *Discrete Mathematics for Computer Science* (undergraduate), and *Computer Graphics* (undergraduate), supporting instruction, assessment, office hours, and student project development.

Complete Publication Record

† Graduate student mentored by Wenyue Hua. Publication and manuscript records cross-checked against DBLP, Microsoft Research, and the author's publication page.

Peer-Reviewed Journal Publications (5)

5. **[TMLR 2026]** Huan-ang Gao, Jiayi Geng, **Wenyue Hua**, Mengkang Hu, Xinzhe Juan, Hongzhang Liu, Shilong Liu, Jiahao Qiu, Xuan Qi, Qihan Ren, Yiran Wu, Hongru Wang, Han Xiao, Yuhang Zhou, Shaokun Zhang, Jiayi Zhang, Jinyu Xiang, Yixiong Fang, Qiwen Zhao, Dongrui Liu, Cheng Qian, Zhenhailong Wang, Minda Hu, Huazheng Wang, Qingyun Wu, Heng Ji, Mengdi Wang. *A Survey of Self-Evolving Agents: What, When, How, and Where to Evolve on the Path to Artificial Super Intelligence*
4. **[ACM TIST 2026]** Jianchao Ji, Zelong Li, Shuyuan Xu, **Wenyue Hua**, Juntao Tan, Haoming Gong, Yongfeng Zhang. *Probing the Symbolic Logical Reasoning Ability of Large Language Models*
3. **[SIGKDD Explorations 2025]** Jianchao Ji, Yutong Chen, Mingyu Jin, Wujiang Xu, **Wenyue Hua**, Yongfeng Zhang. *MoralBench: Moral Evaluation of LLMs*
2. **[Journal of Healthcare Informatics Research 2024]** Huizi Yu, Lizhou Fan, Lingyao Li, Jiayan Zhou, Zihui Ma, Lu Xian, **Wenyue Hua**, Sijia He, Mingyu Jin, Yongfeng Zhang, Ashvin Gandhi, Xin Ma. *Large Language Models in Biomedical and Health Informatics: A Review with Bibliometric Analysis*
1. **[TACL 2023]** **Wenyue Hua**, Lifeng Jin, Linfeng Song, Haitao Mi, Yongfeng Zhang, Dong Yu. *Discover, Explain, Improve: An Automatic Slice Detection Benchmark for Natural Language Processing*

Peer-Reviewed Conference Publications (34)

37. **[EMNLP 2026]** Zekun Li, Shinda Huang, Jiangtian Wang, Nathan Zhang, Antonis Antoniades, **Wenyue Hua**, Kaijie Zhu, Sirui Zeng, William Yang Wang, Xifeng Yan. *SOPBench: Evaluating Language Agents at Following Standard Operating Procedures and Constraints*
36. **[Findings of EMNLP 2026]** Kangqi Ni†, **Wenyue Hua**, Jinbiao Wei, Tianlong Chen. *AdaSpec: Adaptive Routing by Ranking for Speculative Agents*
35. **[ICLR 2026]** Yilin Guan†, Qingfeng Lan, Sun Fei, Dujian Ding, Devang Acharya, Chi Wang, William Yang Wang, **Wenyue Hua**. *Dynamic Speculative Agent Planning*
34. **[SIGIR 2026]** Minghao Guo, Ziyi Ye, Wujiang Xu, Xi Zhu, **Wenyue Hua**, Dimitris N. Metaxas. *Individual Turing Test: A Case Study of LLM-based Simulation Using Longitudinal Personal Data*
33. **[CIKM 2025]** Qingcheng Zeng, Mingyu Jin, Qinkai Yu, Zhenting Wang, **Wenyue Hua**, Zihao Zhou, Guangyan Sun, Yanda Meng, Shiqing Ma, Qifan Wang, Felix Juefei-Xu, Kaize Ding, Fan Yang, Ruixiang Tang, Yongfeng Zhang. *Uncertainty is Fragile: Manipulating Uncertainty in Large Language Models*
32. **[COLM 2025]** Xiao Pu, Michael Saxon, **Wenyue Hua**, William Yang Wang. *THOUGHTTERMINATOR: Benchmarking, Calibrating, and Mitigating Overthinking in Reasoning Models*
31. **[ICSC 2025]** Zhaoqian Xue, Guanhong Liu, Kai Wei, Chong Zhang, Qingcheng Zeng, Songhua Hu, **Wenyue Hua**, Lizhou Fan, Yongfeng Zhang, Lingyao Li. *Toward Equitable Access: Leveraging Crowdsourced Reviews to Investigate Public Perceptions of Health Resource Accessibility*
30. **[Findings of ACL 2025]** Sam Lin†, **Wenyue Hua***, Lingyao Li, Zhenting Wang, Yongfeng Zhang. *ADO: Automatic Data Optimization for Inputs in LLM Prompts*
29. **[Findings of ACL 2025]** **Wenyue Hua***, Kaijie Zhu*, Lingyao Li, Lizhou Fan, Mingyu Jin, Shuhang Lin, Haochen Xue, Zelong Li, Jindong Wang, Yongfeng Zhang. *Disentangling Logic: The Role of Context in Large Language Model Reasoning Capabilities*
28. **[ACL 2025]** Mingyu Jin, Weidi Luo, Sitao Cheng, Xinyi Wang, **Wenyue Hua**, Ruixiang Tang, William Yang Wang, Yongfeng Zhang. *Disentangling Memory and Reasoning Ability in Large Language Models*
27. **[NAACL 2025]** Sam Lin†, **Wenyue Hua***, Zhenting Wang, Mingyu Jin, Lizhou Fan, Yongfeng Zhang. *EmojiPrompt: Generative Prompt Obfuscation for Privacy-Preserving Communication with Cloud-based LLMs*

26. **[COLING 2025]** Mingyu Jin, Qinkai Yu, Jingyuan Huang, Qingcheng Zeng, Zhenting Wang, **Wenyue Hua**, Haiyan Zhao, Kai Mei, Yanda Meng, Kaize Ding, Fan Yang, Mengnan Du, Yongfeng Zhang. *Exploring Concept Depth: How Large Language Models Acquire Knowledge and Concept at Different Layers?*
25. **[ICLR 2025]** Zeru Shi, Kai Mei, Mingyu Jin, Yongye Su, Chaoji Zuo, **Wenyue Hua**, Wujiang Xu, Yujie Ren, Zirui Liu, Mengnan Du, Dong Deng, Yongfeng Zhang. *From Commands to Prompts: LLM-based Semantic File System for AIOS*
24. **[ACL 2025] Wenyue Hua**, Tyler Wong, Fei Sun, Liangming Pan, Adam Jardine, William Yang Wang. *InductionBench: LLMs Fail in the Simplest Complexity Class*
23. **[ICLR 2025] Wenyue Hua**, Mengting Wan, Jagannath Shashank Subramanya Sai Vadrevu, Ryan Nadel, Yongfeng Zhang, Chi Wang. *Interactive Speculative Planning: Enhance Agent Efficiency through Co-design of System and User Interface*
22. **[NAACL 2025]** Yang Ouyang, Hengrui Gu, Shuhang Lin, **Wenyue Hua**, Jie Peng, Bhavya Kailkhura, Meijun Gao, Tianlong Chen, Kaixiong Zhou. *Layer-Level Self-Exposure and Patch: Affirmative Token Mitigation for Jailbreak Attack Defense*
21. **[Findings of ACL 2025]** Jingwen Cheng, Kshitish Ghate, **Wenyue Hua**, William Yang Wang, Hong Shen, Fei Fang. *REALM: A Dataset of Real-World LLM Use Cases*
20. **[ACL 2025]** Ruiwen Zhou†, **Wenyue Hua**, Liangming Pan, Sitao Cheng, Xiaobao Wu, En Yu, William Yang Wang. *RuleArena: A Benchmark for Rule-Guided Reasoning with LLMs in Real-World Scenarios*
19. **[Findings of ACL 2025]** Chaoran Chen, Bingsheng Yao, Ruishi Zou, **Wenyue Hua**, Weimin Lyu, Toby Jia-Jun Li, Dakuo Wang. *Towards a Design Guideline for RPA Evaluation: A Survey of Large Language Model-Based Role-Playing Agents*
18. **[CIKM 2025]** Qingcheng Zeng, Mingyu Jin, Qinkai Yu, Zhenting Wang, **Wenyue Hua**, Guangyan Sun, Yanda Meng, Shiqing Ma, Qifan Wang, Felix Juefei-Xu, Fan Yang, Kaize Ding, Ruixiang Tang, Yongfeng Zhang. *Uncertainty Quantification for Multiple-Choice Questions is Just One-Token Deep*
17. **[EMNLP System Demonstrations 2024]** Shuhang Lin*, **Wenyue Hua***, Lingyao Li, Che-Jui Chang, Lizhou Fan, Jianchao Ji, Hang Hua, Mingyu Jin, Jiebo Luo, Yongfeng Zhang. *BattleAgent: Multi-modal Dynamic Emulation on Historical Battles to Complement Historical Analysis*
16. **[ECIR 2024]** Jianchao Ji, Zelong Li, Shuyuan Xu, **Wenyue Hua**, Yingqiang Ge, Juntao Tan, Yongfeng Zhang. *GenRec: Large Language Model for Generative Recommendation*
15. **[SIGIR 2024]** Juntao Tan, Shuyuan Xu, **Wenyue Hua**, Yingqiang Ge, Zelong Li, Yongfeng Zhang. *IDGenRec: LLM-RecSys Alignment with Textual ID Learning*
14. **[Findings of EMNLP 2024]** Alfonso Amayuelas, Xianjun Yang, Antonis Antoniades, **Wenyue Hua**, Liangming Pan, William Yang Wang. *MultiAgent Collaboration Attack: Investigating Adversarial Attacks in Large Language Model Collaborations via Debate*
13. **[ACL 2024]** Lizhou Fan*, **Wenyue Hua***, Lingyao Li, Haoyang Ling, Yongfeng Zhang. *NPHardEval: Dynamic Benchmark on Reasoning Ability of Large Language Models via Complexity Classes*
12. **[SIGIR 2024]** Shuyuan Xu, **Wenyue Hua**, Yongfeng Zhang. *OpenP5: An Open-Source Platform for Developing, Training, and Evaluating LLM-based Recommender Systems*
11. **[Findings of ACL 2024] Wenyue Hua***, Jiang Guo*, Mingwen Dong, Henghui Zhu, Patrick Ng, Zhiguo Wang. *Propagation and Pitfalls: Reasoning-based Assessment of Knowledge Editing through Counterfactual Tasks*
10. **[Findings of ACL 2024]** Mingyu Jin, Qinkai Yu, Dong Shu, Haiyan Zhao, **Wenyue Hua**, Yanda Meng, Yongfeng Zhang, Mengnan Du. *The Impact of Reasoning Step Length on Large Language Models*
9. **[Findings of EMNLP 2024] Wenyue Hua**, Xianjun Yang, Mingyu Jin, Zelong Li, Wei Cheng, Ruixiang Tang, Yongfeng Zhang. *TrustAgent: Towards Safe and Trustworthy LLM-based Agents*
8. **[EACL 2024] Wenyue Hua**, Yingqiang Ge, Shuyuan Xu, Jianchao Ji, Zelong Li, Yongfeng Zhang. *UP5: Unbiased Foundation Model for Fairness-aware Recommendation*
7. **[SIGIR-AP 2023] Wenyue Hua**, Shuyuan Xu, Yingqiang Ge, Yongfeng Zhang. *How to Index Item IDs for Recommendation Foundation Models*
6. **[NeurIPS 2023]** Yingqiang Ge, **Wenyue Hua**, Kai Mei, Jianchao Ji, Juntao Tan, Shuyuan Xu, Zelong Li, Yongfeng Zhang. *OpenAGI: When LLM Meets Domain Experts*
5. **[RecSys 2023] Wenyue Hua**, Lei Li, Shuyuan Xu, Li Chen, Yongfeng Zhang. *Tutorial on Large Language Models for Recommendation*
4. **[ICLR 2022]** Wenzheng Zhang, **Wenyue Hua**, Karl Stratos. *EntQA: Entity Linking as Question Answering*
3. **[Findings of EMNLP 2022] Wenyue Hua**, Yongfeng Zhang. *System 1 + System 2 = Better World: Neural-Symbolic Chain of Logic Reasoning*
2. **[ICGI 2021] Wenyue Hua**, Adam Jardine. *Learning Input Strictly Local Functions From Their Composition*
1. **[EMNLP 2020]** Mingming Sun, **Wenyue Hua**, Zoey Liu, Xin Wang, Kangjie Zheng, Ping Li. *A Predicate-Function-Argument Annotation of Natural Language for Open-Domain Information eXpression*

Preprints and Technical Reports (29)

29. **[Preprint 2026]** Wenyue Hua, Zachary Huang, Tyler Payne, Safoora Yousefi, Saleema Amershi, Asli Celikyilmaz. *From Passive Delegates to Strategic Negotiators: Reinforcing Social Reasoning in Small Language Models with SocialRL*
28. **[Technical Report 2026] Wenyue Hua**, Sripad Karne, Qian Xie, Armaan Agrawal, Nikos Pagonas, Kostis Kaffes, Tianyi Peng. *AgentOpt v0.1 Technical Report: Client-Side Optimization for LLM-Based Agent*
27. **[Preprint 2026]** Chengzhi Liu, Yichen Guo, Yepeng Liu, Yuzhe Yang, Qianqi Yan, Xuandong Zhao, **Wenyue Hua**, Sheng Liu, Sharon Li, Yuheng Bu, Xin Eric Wang. *Auditing Agent Harness Safety*
26. **[Preprint 2026]** Kangqi Ni†, **Wenyue Hua**, Xiaoxiang Shi, Jiang Guo, Shiyu Chang, Tianlong Chen. *Chimera: Latency- and Performance-Aware Multi-agent Serving for Heterogeneous LLMs*

25. **[Preprint 2026]** Ruiwen Zhou†, Maojia Song, Xiaobao Wu, Sitao Cheng, Xunjian Yin, Yuxi Xie, Zhuoqun Hao, **Wenyue Hua**, Liangming Pan, Soujanya Poria, Min-Yen Kan. *Epistemic Context Learning: Building Trust the Right Way in LLM-Based Multi-Agent Systems*
24. **[Preprint 2026]** Keyang Xuan, Peiyang Song, Pan Lu, Pengrui Han, Wenkai Li, Zhenyu Zhang, Zexue He, **Wenyue Hua**, Manling Li, Jiaxuan You, Adrian Weller, Yizhong Wang, Jiaxin Pei. *Interactive Evaluation Requires a Design Science*
23. **[Preprint 2026]** Wujiang Xu, Yu Wang, Kai Mei, Kaiqu Liang, Zhenting Wang, Mingyu Jin, Han Zhang, Shi-Xiong Zhang, **Wenyue Hua**, Sambit Sahu, Dimitris N. Metaxas. *MemGym: a Long-Horizon Memory Environment for LLM Agents*
22. **[Preprint 2026]** **Wenyue Hua**, Tianyi Peng, Chi Wang, Ian Kaufman, Bryan Lim, Chandler Fang. *Quantifying Trust: Financial Risk Management for Trustworthy AI Agents*
21. **[Preprint 2025]** Zihui Ma, Lingyao Li, Juan Li, **Wenyue Hua**, Jingxiao Liu, Qingyuan Feng, Yuki Miura. *A Multimodal, Multilingual, and Multidimensional Pipeline for Fine-grained Crowdsourcing Earthquake Damage Evaluation*
20. **[Preprint 2025]** Shurui Cao, **Wenyue Hua**, William Yang Wang, Hong Shen, Fei Fang. *Can Online GenAI Discussion Serve as Bellwether for Labor Market Shifts?*
19. **[Preprint 2025]** Xiang Li, Huizi Yu, Wenkong Wang, Yiran Wu, Jiayan Zhou, **Wenyue Hua**, Xinxin Lin, Wenjia Tan, Lexuan Zhu, Bingyi Chen, Guang Chen, Ming-Li Chen, Yang Zhou, Zhao Li, Themistocles L. Assimes, Yongfeng Zhang, Qingyun Wu, Xin Ma, Lingyao Li, Lizhou Fan. *DispatchMAS: Fusing taxonomy and artificial intelligence agents for emergency medical services*
18. **[Preprint 2025]** Gurusha Juneja†, Jayanth Naga Sai Pasupulati, Alon Albalak, **Wenyue Hua**, William Yang Wang. *MAGPIE: A benchmark for Multi-AGent contextual PrIvacy Evaluation*
17. **[Microsoft Research Technical Report 2025]** Gagan Bansal, **Wenyue Hua**, Zezhou Huang, Adam Fourney, Amanda Swearngin, Will Epperson, Tyler Payne, Jake M. Hofman, Brendan Lucier, Chinmay Singh, Markus Mobius, Akshay Nambi, Archana Yadav, Kevin Gao, David M. Rothschild, Aleksandrs Slivkins, Daniel G. Goldstein, Hussein Mozannar, Nicole Immorlica, Maya Murad, Matthew Vogel, Subbarao Kambhampati, Eric Horvitz, Saleema Amershi. *Magentic Marketplace: An Open-Source Environment for Studying Agentic Markets*
16. **[Preprint 2025]** Lingyao Li, Haolun Wu, Zhenkun Li, Jiabei Hu, Yu Wang, Xiaoshan Huang, **Wenyue Hua**, Wenqian Wang. *PartnerMAS: An LLM Hierarchical Multi-Agent Framework for Business Partner Selection on High-Dimensional Features*
15. **[Preprint 2025]** **Wenyue Hua**, Dujian Ding, Yile Gu, Yujie Ren, Kai Mei, Minghua Ma, William Yang Wang. *Semantic Scheduling for LLM Inference*
14. **[Preprint 2025]** Yiwen Zhang, Yifu Wu, **Wenyue Hua**, Xuming Hu. *Understanding Dynamic Diffusion Process of LLM-based Agents under Information Asymmetry*
13. **[Preprint 2024]** Lingyao Li, Jiayan Zhou, Zhenxiang Gao, **Wenyue Hua**, Lizhou Fan, Huizi Yu, Loni Hagen, Yongfeng Zhang, Themistocles L. Assimes, Libby Hemphill, Siyuan Ma. *A scoping review of using Large Language Models (LLMs) to investigate Electronic Health Records (EHRs)*
12. **[Preprint 2024]** Huizi Yu, Jiayan Zhou, Lingyao Li, Shan Chen, Jack Gallifant, Anye Shi, Xiang Li, **Wenyue Hua**, Mingyu Jin, Guang Chen, Yang Zhou, Zhao Li, Trisha Gupte, Ming-Li Chen, Zahra Azizi, Yongfeng Zhang, Themistocles L. Assimes, Xin Ma, Danielle S. Bitterman, Lin Lu, Lizhou Fan. *AIPatient: Simulating Patients with EHRs and LLM Powered Agentic Workflow*
11. **[Preprint 2024]** Zelong Li, Shuyuan Xu, Kai Mei, **Wenyue Hua**, Balaji Rama, Om Raheja, Hao Wang, He Zhu, Yongfeng Zhang. *AutoFlow: Automated Workflow Generation for Large Language Model Agents*
10. **[Preprint 2024]** Zelong Li, **Wenyue Hua**, Hao Wang, He Zhu, Yongfeng Zhang. *Formal-LLM: Integrating Formal Language and Natural Language for Controllable LLM-based Agents*
9. **[Preprint 2024]** **Wenyue Hua**, Ollie Liu, Lingyao Li, Alfonso Amayuelas, Julie Chen, Lucas Jiang, Mingyu Jin, Lizhou Fan, Fei Sun, William Wang, Xintong Wang, Yongfeng Zhang. *Game-theoretic LLM: Agent Workflow for Negotiation Games*
8. **[Preprint 2024]** Mingyu Jin, Qinkai Yu, Dong Shu, Chong Zhang, Lizhou Fan, **Wenyue Hua**, Suiyuan Zhu, Yanda Meng, Zhenting Wang, Mengnan Du, Yongfeng Zhang, Yanda Meng. *Health-LLM: Personalized Retrieval-Augmented Disease Prediction System*
7. **[Preprint 2024]** Lizhou Fan, **Wenyue Hua**, Xiang Li, Kaijie Zhu, Mingyu Jin, Lingyao Li, Haoyang Ling, Jinkui Chi, Jindong Wang, Xin Ma, Yongfeng Zhang. *NPHardEval4V: A Dynamic Reasoning Benchmark of Multimodal Large Language Models*
6. **[Preprint 2024]** Zelong Li, Jianchao Ji, Yingqiang Ge, **Wenyue Hua**, Yongfeng Zhang. *PAP-REC: Personalized Automatic Prompt for Recommendation Language Model*
5. **[Preprint 2024]** Mingyu Jin, Beichen Wang, Zhaoqian Xue, Suiyuan Zhu, **Wenyue Hua**, Hua Tang, Kai Mei, Mengnan Du, Yongfeng Zhang. *What if LLMs Have Different World Views: Simulating Alien Civilizations with LLM-based Agents*
4. **[Preprint 2024]** Chong Zhang, Xinyi Liu, Mingyu Jin, Zhongmou Zhang, Lingyao Li, Zhenting Wang, **Wenyue Hua**, Dong Shu, Suiyuan Zhu, Xiaobo Jin, Sujian Li, Mengnan Du, Yongfeng Zhang. *When AI Meets Finance (StockAgent): Large Language Model-based Stock Trading in Simulated Real-world Environments*
3. **[Preprint 2023]** Yingqiang Ge, Yujie Ren, **Wenyue Hua**, Shuyuan Xu, Juntao Tan, Yongfeng Zhang. *LLM as OS, Agents as Apps: Envisioning AIOS, Agents and the AIOS-Agent Ecosystem*
2. **[Preprint 2023]** **Wenyue Hua**, Lizhou Fan, Lingyao Li, Kai Mei, Jianchao Ji, Yingqiang Ge, Libby Hemphill, Yongfeng Zhang. *War and Peace (WarAgent): Large Language Model-based Multi-Agent Simulation of World Wars*
1. **[Preprint 2022]** **Wenyue Hua**, Yuchen Zhang, Zhe Chen, Josie Li, Melanie Weber. *LegalRelectra: Mixed-domain Language Modeling for Long-range Legal Text Comprehension*

Other Scholarly Works (3)

3. **[M.A. thesis, Rutgers University 2020]** **Wenyue Hua**. *Learning Underlying Representations and Input-Strictly-Local Functions*

2. **[Scholarly paper 2019] Wenyue Hua**. *Computational Complexity in Optional Syllabification of Yavapai*
1. **[Scholarly paper 2019] Wenyue Hua**. *Free Choice Disjunction of Propositions in Generic Sentences*

Total listed: 74 distinct peer-reviewed publications, manuscripts, technical reports, and scholarly works.

Invited Talks and Seminars

From Philosophy of Language to AI Agents. Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi. — Jun. 2026

From Philosophy of Language to AI Agents. University of California, Santa Barbara, Department of Computer Science. — Jun. 2026

Agentic Risk Standard. Columbia University, DAPLab / Department of Computer Science, New York. — May 2026

Agentic Risk Standard. Carnegie Mellon University, Department of Computer Science. — May 2026

Agentic Risk Standard. University of California, Santa Barbara, Department of Computer Science. — Apr. 2026

From Philosophy of Language to AI Agents. Fermilab. — Mar. 2026

Magentic Marketplace. Columbia Agent Workshop, Columbia University. — Oct. 2025

Magentic Marketplace. RecSys 2025 Workshop on Evaluating and Applying Recommendations with LLMs (EARL). — Sep. 2025

InductionBench: LLMs Fail in the Simplest Complexity Class. Formal Languages and Neural Networks (FLaNN) Seminar. — May 2025

Building LLM Agents for Safety and Efficiency. Northeastern University. — Mar. 2025

LLM-Based Agents for Recommendation Systems. Amazon. — Feb. 2025

Interactive Speculative Planning. AG2. — Feb. 2025

Game-Theoretic LLM: Agent Workflow for Negotiation Games. Vanderbilt University. — Jan. 2025

LLM Agents for Decision-Making in International Relations and Game Theory. Beijing Institute of Mathematical Sciences and Applications (BIMSA). — Nov. 2024

WarAgent: Multi-Agent Simulation of World Wars. European Social Simulation Association Annual Conference. — May 2024

Professional Service and Workshop Organization

Guest Editor. TIST Special Issue on Knowledge-Informed Large Language Models: Integrating Physical Laws, Symbolic Reasoning, and Data-Driven Learning, Apr. 27, 2026. — 2026

Co-organizer. ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents), Apr. 27, 2026. — 2026

Co-organizer. The 2nd Workshop on Lifelong Agents: Learning, Aligning, and Evolving, COLM 2026, San Francisco; forthcoming Oct. 9, 2026. — 2026

Area Chair. ACL, EMNLP, NAACL, EACL, COLM — 2025 - 2026

Reviewer. ICLR, NeurIPS, ICML — 2022 - 2026

Founder. NICE AI TALK, supporting research exchange and invited presentations in AI. — Ongoing

Industry Intern Experience

Microsoft Research, AutoGen May 2024 – Aug. 2024
Research Intern *Redmond, WA*

- Introduced speculative agent planning, reducing end-to-end planning latency by 40%.
- Designed scheduling and interface mechanisms that support active user intervention while preserving latency gains.

Amazon, AWS AI Lab / QuickSight Q May 2023 – Aug. 2023
Applied Scientist Intern *United States*

- Built a seven-category knowledge-editing benchmark for discrete reasoning and evaluated input augmentation, QLoRA, MEMIT, and MEND across model sizes and reasoning types.

Tencent America, AI Lab May 2022 – Aug. 2022
Research Scientist Intern *United States*

- Developed automatic error-detection models from binary classification to multi-label and sequence-to-sequence settings, with evaluation metrics and statistical explanations for discovered error slices.

Research Funding, Awards, and Honors

KAUST AI Rising Star	2025
National Science Foundation SBIR research funding ($50,000)	2021
Phi Beta Kappa; UCLA College Honors and Departmental Honors	2018
UCLA Dean's Honors List	2014–2018

Technical Skills

Research	LLM agents; trustworthy AI; natural language processing and text mining; computational social science; multi-agent reinforcement learning; post-training (SFT, PPO, distillation); privacy-preserving alignment; model routing and selection; serving, scheduling, and interactive evaluation.
Programming	Python, C++, PyTorch, vLLM, FlashAttention, Linux, Git, LaTeX, Haskell, MATLAB, R.